HOW THE UI ANSWERS "WHO OR WHAT CAN ACCESS?" WITH CEDAR

The Customer Collaboration Cloud's user interface often needs to display visibility information such as lists of documents a person can view or the people who have access to a document. These are discovery questions, not enforcement checks. Cedar itself cannot perform searches; it evaluates access requests. To efficiently generate these UI lists, C³ combines its ReBAC graph and policy engine in a three-stage process:

1. Get candidates from the relationship graph

Use the tenant's ReBAC graph, which records relationships like owner, editor, viewer, member-of-team, and team-has-access-to-folder to quickly retrieve a set of candidate principals or resources.

- To find *docs this user can access*, start at the principal node and move outward along relationships to identify the resources they can reach.
- To find *users who can access this doc*, start at the resource node and move outward along inbound relationships (from the resource's perspective) to identify connected principals.

The direction depends on where the query begins. Both searches use the same relationship graph. The traversal direction just reverses depending on whether you're starting from a user or a document.

2. Apply a policy scope prefilter

Some policy conditions never change at runtime, such as document labels, classification, or the tenant a resource belongs to.

When a policy is published, C³ *partially evaluates* it using all the information that's already known (such as action and tenant). The result is a simplified predicate or scope that describes which resources or principals could ever satisfy the policy.

Partial evaluation is an experimental feature in recent Cedar versions that returns the simplified predicate, known as a policy residual. Chapter 17